ZCAS University

CCS3631 - MOBILE AND WIRELESS FUNDAMENTALS

MID-TERM TEST

20TH APRIL 2026

15:30 -17:00 HOURS

TIME ALLOWED: ONE AND A HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Choose any of the **TWO** questions from the three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

QUESTION 1

a) A mobile network operator discovers that several subscribers are receiving services without being billed, suggesting a fraudulent bypass of the authentication system. Describe the steps the operator should take to identify the source of the fraud, restore billing integrity, and prevent future occurrences using GSM security mechanisms. [Marks 6]

b) A GSM network experiences a situation where a mobile subscriber moving between cities loses their call connection each time they cross from one Mobile Switching Centre (MSC) coverage area to another. Explain how the GSM network should handle this scenario to maintain seamless communication, and identify the type of handover involved. [Marks 7]

c) An attacker intercepts radio transmissions between a mobile station and a Base Transceiver Station (BTS), attempting to eavesdrop on subscriber calls and steal identity information. Describe the GSM security mechanisms that are in place to protect against such an attack, and explain how each mechanism works to safeguard user data. [Marks 6]

d) What are the four subsystems that make up the structure of a GSM network, and what is the primary function of each? [Marks 6]

MARKS 25

QUESTION 2

a) A telecommunications company is rolling out a new GSM network in a rural area where coverage is limited. Subscribers complain that their calls drop frequently as they travel between villages. Using your knowledge of GSM handover types, explain what technical measures the network should implement to ensure uninterrupted communication, and describe which type(s) of handover would be most relevant in this scenario. [Marks 6]

b) A subscriber travels internationally and expects to use their mobile phone in a foreign country. However, they are unable to make or receive calls upon arrival. Using your understanding of GSM network architecture and security, explain the possible technical reasons for this failure and describe how roaming agreements and authentication mechanisms should work to enable cross-border service. [Marks 6]

c) A GSM network operator suspects that cloned SIM cards are being used to access their network without authorisation. Describe how the GSM authentication process, including

the use of IMSI, RAND, SRES, and the A3 algorithm, works to detect and prevent such unauthorised access. [Marks 8]

d) Distinguish between the roles of the Base Transceiver Station (BTS) and the Base Station Controller (BSC) within the Base Station Subsystem (BSS). [Marks 5]

MARKS 25

QUESTION 3

a) Briefly explain the role of the Home Location Register (HLR) and Visitor Location Register (VLR) in the GSM Network Switching Subsystem (NSS). [Marks 4]

b) What is the purpose of the Operation and Support Subsystem (OSS) in a GSM network, and who is responsible for monitoring it? [Marks 4]

c) Explain what TDMA is and describe how GSM uses it as part of its communication standard. [Marks 4]

d) What are the three interfaces used to connect the four subsystems of a GSM network, and what components does each interface link? [Marks 6]

e) State two security concerns from the operator's perspective and two from the customer's perspective in a GSM network. [Marks 4]

f) What is the significance of the Temporary Mobile Subscriber Identity (TMSI), and how does it enhance subscriber confidentiality in a GSM network? [Marks 3]

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